



## AQ6.1: Activity Questions 1 - Not Graded



This assignment will not be graded and is only for practice.

### Level 1

1 point

Suppose  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is a linear transformation, such that  $T(x, y) = (x, 0)$ . Which of the following options are correct?

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The matrix corresponding to  $T$  with respect to the standard ordered basis of  $\mathbb{R}^2$ , i.e.,  $\{(1, 0), (0, 1)\}$ , for both the domain and co-domain is  $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ .

☐

The matrix corresponding to  $T$  with respect to the standard ordered basis of  $\mathbb{R}^2$ , i.e.,  $\{(1, 0), (0, 1)\}$ , for both the domain and co-domain is  $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ .

☐

$T$  is neither one-one nor onto.

☐

$T$  is one-one but not onto.

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

The matrix corresponding to  $T$  with respect to the standard ordered basis of  $\mathbb{R}^2$ , i.e.,  $\{(1, 0), (0, 1)\}$ , for both the domain and co-domain is  $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ .

$T$  is neither one-one nor onto.

1 point

If the matrix corresponding to a linear transformation  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ , with respect to standard ordered basis of  $\mathbb{R}^2$ , i.e.,  $\{(1, 0), (0, 1)\}$ , for both the domain and co-domain, is  $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$ . Which of the following is the appropriate definition of  $T$ ?

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$T(x, y) = (2x + y, 3x + 4y)$

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$T(x, y) = (2x + y, 3x - 4y)$

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$T(x, y) = (x + 4y, 2x + 3y)$

☐

$T(x, y) = (2x + 3y, x + 4y)$

**No, the answer is incorrect.**

**Score: 0**

### Accepted Answers:

$T(x, y) = (2x + 3y, x + 4y)$

Let  $W = \{(x, y, z) \mid x = 2y + z\}$  be a subspace of  $\mathbb{R}^3$ . Let

$\beta = \{(2, 1, 0), (1, 0, 1)\}$  be a basis of  $W$ . Let  $T: W \rightarrow \mathbb{R}^2$  be a linear transformation such that  $T(2, 1, 0) = (1, 0)$  and  $T(1, 0, 1) = (0, 1)$ . Answer the questions 3, 4 and 5 using the given information.

1 point



Which of the following is the appropriate definition of  $T$ ?

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$$T(x, y, z) = (x, y) \quad T(x, y, z) = (x, y)$$

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$$T(x, y, z) = (x, z) \quad T(x, y, z) = (x, z)$$

☐

$$T(x, y, z) = (y, z) \quad T(x, y, z) = (y, z)$$

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$$T(x, y, z) = (x - y - z, x - 2y) \quad T(x, y, z) = (x - y - z, x - 2y)$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$T(x, y, z) = (y, z) \quad T(x, y, z) = (y, z)$$

$$T(x, y, z) = (x - y - z, x - 2y) \quad T(x, y, z) = (x - y - z, x - 2y)$$

1 point

Choose the correct options.

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$T$  is one to one but not onto.

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$T$  is onto but not one to one.

☐

$T$  is neither one to one nor onto.

☐

$T$  is an isomorphism.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$T$  is an isomorphism.

1 point

What will be the matrix representation of  $T$  with respect to the basis  $\beta$  for  $W$  and  $\gamma = \{(1, 1), (1, -1)\}$  for  $R^2$ ?

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$$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} {}^{21} [111-1]$$

☐

$$\frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} {}^{21} [1001]$$

☐

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} [111-1]$$

☐

$$\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} [1-111]$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} {}^{21} [111-1]$$

1 point

Choose the set of correct options.

☐

Let  $u = (3, 1, 0)$ ,  $v = (0, 1, 7)$ , and  $w = (3, 0, -7)$ . There is a linear

transformation  $T: R^3 \rightarrow R^3$  such that  $T(u) = T(v) = (0, 0, 0)$  and that  $T(w) = (5, 1, 0)$ .

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Let  $u = (2, 1, 0)$ ,  $v = (1, 0, 1)$ , and  $w = (3, 5, 6)$ . There is a linear transformation  $T: R^3 \rightarrow R^2$  such that  $T(u) = (1, 0)$ ,  $T(v) = (0, 1)$  and that  $T(w) = (5, 6)$ .

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Let  $u = (1, 0, 0)$ ,  $v = (1, 0, 1)$ , and  $w = (0, 1, 0)$ . There is a linear transformation  $T: R^3 \rightarrow R^3$  such that  $T(u) = (0, 0, 0)$ ,  $T(v) = (0, 0, 0)$ ,  $T(w) = (0, 0, 0)$ , and  $T(1, 4, 2) = (2, 4, 1)$ .

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Let  $u = (1, 0)$ , and  $v = (0, 1)$ . There is a linear transformation  $T: R^2 \rightarrow R^2$  such that  $T(u) = (\pi, 1)$ , and  $T(v) = (1, e)$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Let  $u = (2, 1, 0)$ ,  $v = (1, 0, 1)$ , and  $w = (3, 5, 6)$ . There is a linear transformation  $T: R^3 \rightarrow R^2$  such that  $T(u) = (1, 0)$ ,  $T(v) = (0, 1)$  and that  $T(w) = (5, 6)$ .

Let  $u = (1, 0)$ , and  $v = (0, 1)$ . There is a linear transformation  $T: R^2 \rightarrow R^2$  such that  $T(u) = (\pi, 1)$ , and  $T(v) = (1, e)$ .

**Level 2**

*1 point*

Choose the set of correct options.

☐

If the Identity matrix of order 2 is the matrix representation of a linear transformation  $T: R^2 \rightarrow R^2$ , with respect to the standard ordered basis for both the domain and co-domain, then it is also the matrix representation of  $T$  with respect to any other basis  $\beta$  for both the domain and co-domain.

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Let  $T$  be an isomorphism between two vector spaces, and  $A$  be the matrix representation of  $T$  with respect to some basis. Then the nullity of  $A$  is 0.

☐

If  $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$  is the matrix representation of  $T: R^2 \rightarrow R^2$  with respect to the standard ordered basis of  $R^2$ , for both the domain and co-domain, then the matrix representation of  $T \circ T \circ T$  with respect to the same bases will also be  $A$ .

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If  $A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$  is the matrix representation of  $T: R^2 \rightarrow R^2$  with respect to the standard ordered basis of  $R^2$ , for both the domain and co-domain, then the matrix representation of  $T \circ T \circ T$  will be the identity matrix.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If the Identity matrix of order 2 is the matrix representation of a linear transformation  $T: R^2 \rightarrow R^2$ , with respect to the standard ordered basis for both the domain and co-domain, then it is also the matrix representation of  $T$  with respect to any other basis  $\beta$  for both the domain and co-domain.

Let  $T$  be an isomorphism between two vector spaces, and  $A$  be the matrix representation of  $T$

with respect to some basis. Then the nullity of  $AA$  is 0.

If  $AA = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$  is the matrix representation of  $T: R^2 \rightarrow R^2$  with respect to the

standard ordered basis of  $R^2$ , for both the domain and co-domain, then the matrix representation of  $T \circ T \circ T$  with respect to the same bases will also be  $AA$ .

Let  $\beta = \{(1, 0), (0, 1)\}$  and  $\gamma = \{(1, 1), (1, -1)\}$  be two bases of  $R^2$ . If  $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$  is the matrix representation of a linear transformation  $T: R^2 \rightarrow R^2$

with respect to  $\beta$  for both the domain and the co-domain. Answer questions 8 and 9 using the information given above.

1 point

What is the matrix representation of  $T$  with respect to  $\gamma$  for the domain and  $\beta$  for the co-domain?

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$$\begin{bmatrix} a + b & c + d \\ a - b & c - d \end{bmatrix} [a + ba - bc + dc - d]$$

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$$\begin{bmatrix} a - b \\ c - d \end{bmatrix} [ac - b - d]$$

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$$\begin{bmatrix} a + b & a - b \\ c + d & c - d \end{bmatrix} [a + bc + da - bc - d]$$

☐

$$\begin{bmatrix} a + b & a - b \\ c + d & c - d \end{bmatrix} [a + bc + d - a - b - c - d]$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\begin{bmatrix} a + b & a - b \\ c + d & c - d \end{bmatrix} [a + bc + da - bc - d]$$

1 point

What is the matrix representation of  $T$  with respect to  $\gamma$  for both the domain and the co-domain?

☐

$$\begin{bmatrix} a + b + c + d & a - b + c - d \\ a - b - c - d & a - b - c + d \end{bmatrix} [a + b + c + da + b - c - da - b + c - da - b - c + d]$$

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$$\frac{a + b + c + da - b + c - d}{2} \frac{a - b - c - da - b - c + d}{2} [2a + b + c + d \ 2a + b - c - d \ 2a - b + c - d \ 2a - b - c + d]$$

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$$\frac{a + b + c + da + b - c - d}{2} \frac{a - b - c - da - b - c + d}{2} [2a + b + c + d \ 2a - b + c - d \ 2a + b - c - d \ 2a - b - c + d]$$

☐

$$\begin{bmatrix} a + b + c + d & a - b + c - d \\ a - b - c - d & a - b - c + d \end{bmatrix} [a + b + c + da - b + c - da + b - c - da - b - c + d]$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\frac{a + b + c + da - b + c - d}{2} \frac{a - b - c - da - b - c + d}{2} [2a + b + c + d \ 2a + b - c - d \ 2a - b + c - d \ 2a - b - c + d]$$

1 point

Which of the following options are correct?

☐

If  $T: R^2 \rightarrow R^2$  is a linear transformation which is both one-one and onto then  $T^{-1}T = I$

is also a linear transformation.



If  $T: R^n \rightarrow R^m$   $T: R_n \rightarrow R_m$  then the matrix corresponding to  $T^T$ , with respect to the standard bases for  $R^n$   $R_n$  and  $R^m$   $R_m$  is of order  $n \times m$ .



If  $T: R^n \rightarrow R^m$   $T: R_n \rightarrow R_m$  then the matrix corresponding to  $T^T$ , with respect to the standard bases for  $R^n$   $R_n$  and  $R^m$   $R_m$  is of order  $m \times n$ .



Consider the vector space  $V = \left\{ \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid a, b, c \in R, \right\}$   $V = \{[a0bc] \mid a, b, c \in R, \}$ , that is, the set of all  $2 \times 2$  upper triangular matrices with usual matrix addition and scalar multiplication. Then there exists an isomorphism from  $R^3$  to  $V$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

If  $T: R^2 \rightarrow R^2$   $T: R_2 \rightarrow R_2$  is a linear transformation which is both one-one and onto then  $T^{-1}$  is also a linear transformation.

If  $T: R^n \rightarrow R^m$   $T: R_n \rightarrow R_m$  then the matrix corresponding to  $T^T$ , with respect to the standard bases for  $R^n$   $R_n$  and  $R^m$   $R_m$  is of order  $m \times n$ .

Consider the vector space  $V = \left\{ \begin{bmatrix} a & b \\ 0 & c \end{bmatrix} \mid a, b, c \in R, \right\}$   $V = \{[a0bc] \mid a, b, c \in R, \}$ , that is, the set of all  $2 \times 2$  upper triangular matrices with usual matrix addition and scalar multiplication. Then there exists an isomorphism from  $R^3$  to  $V$ .

[Check Answers](#)

Your score is: 0/10